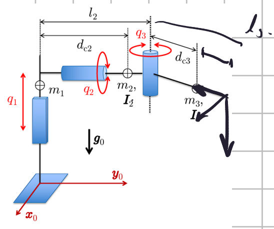
Exercise 1

Consider the robot in Fig. 1 with $n = 3$ joints, the first one prismatic and the other two revolute. Each link has uniformly distributed mass, center of mass on its major physical axis, and a diagonal barycentric inertia matrix. Assume that friction at the joints can be neglected. The robot is commanded at the joint level by a generalized vector of forces/torques $\boldsymbol{\tau} \in \mathbb{R}^3$.

a) Derive the dynamic model of the robot in the Lagrangian form $\boldsymbol{M}(\boldsymbol{q})\ddot{\boldsymbol{q}} + \boldsymbol{c}(\boldsymbol{q}, \dot{\boldsymbol{q}}) + \boldsymbol{g}(\boldsymbol{q}) = \boldsymbol{\tau}$.

b) Find a linear parametrization $\boldsymbol{Y}(\boldsymbol{q}, \dot{\boldsymbol{q}}, \ddot{\boldsymbol{q}})\,\boldsymbol{a} = \boldsymbol{\tau}$ of the robot dynamics in terms of a vector $\boldsymbol{a} \in \mathbb{R}^p$ of dynamic coefficients and of a $3 \times p$ regressor matrix $\boldsymbol{Y}$. Discuss the minimality of p.

c) Determine which of the $10n = 30$ standard dynamic parameters of the links are irrelevant for the describing the motion of the robot.



α	a	d	θ
$-\pi/2$	0	q_1	0
$-\pi/2$	0	l_2	q_2
0	l_3	0	q_3

$$^0T_1 = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & -1 & 0 & q_1 \\ 0 & 0 & 0 & 1 \end{pmatrix} \qquad {}^1T_2 = \begin{pmatrix} c_2 & 0 & -s_2 & 0 \\ s_2 & 0 & c_2 & 0 \\ 0 & -1 & 0 & l_2 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

$$^1\omega_1 = 0$$

$$^0T_3 = \begin{pmatrix} c_3 & -s_3 & 0 & l_3c_3 \\ s_3 & c_3 & 0 & l_3s_3 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \qquad {}^1v_1 = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & -1 \\ 0 & 1 & 0 \end{pmatrix}\begin{pmatrix} 0 \\ 0 \\ \dot{q}_1 \end{pmatrix} = \begin{pmatrix} 0 \\ -\dot{q}_1 \\ 0 \end{pmatrix} \qquad v_{c1} = (0 \;\; -\dot{q}_1 \;\; 0) \Rightarrow T_1 = \frac{1}{2} m_1 \dot{q}_1^2$$

$$^2\omega_2 = \begin{pmatrix} c_2 & s_2 & 0 \\ 0 & 0 & -1 \\ -s_2 & c_2 & 0 \end{pmatrix}\begin{pmatrix} 0 \\ 0 \\ \dot{q}_2 \end{pmatrix} = (0 \;\; -\dot{q}_2 \;\; 0)^T \qquad {}^2v_2 = \begin{pmatrix} c_2 & s_2 & 0 \\ 0 & 0 & -1 \\ -s_2 & c_2 & 0 \end{pmatrix}\left[\begin{pmatrix} 0 \\ -\dot{q}_1 \\ 0 \end{pmatrix} + \begin{pmatrix} 0 & -\dot{q}_2 & 0 \\ \dot{q}_2 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}\begin{pmatrix} 0 \\ 0 \\ l_2 \end{pmatrix}\right] = \begin{pmatrix} -\dot{q}_1 s_2 \\ 0 \\ -\dot{q}_1 c_2 \end{pmatrix}$$

$$v_{c2} = \begin{pmatrix} -\dot{q}_1 s_2 \\ 0 \\ -\dot{q}_1 c_2 \end{pmatrix} + \begin{pmatrix} 0 \\ -\dot{q}_2 \\ 0 \end{pmatrix} \times \begin{pmatrix} 0 \\ d_{c2} - l_2 \\ 0 \end{pmatrix} = \begin{pmatrix} -\dot{q}_1 s_2 \\ 0 \\ -\dot{q}_1 c_2 \end{pmatrix} \Rightarrow T_2 = \frac{1}{2} m_2 \dot{q}_1^2 + \frac{1}{2} I_{2y} \dot{q}_2^2$$

$$^3\omega_3 = \begin{pmatrix} c_3 & s_3 & 0 \\ -s_3 & c_3 & 0 \\ 0 & 0 & 1 \end{pmatrix}\begin{pmatrix} 0 \\ -\dot{q}_2 \\ \dot{q}_3 \end{pmatrix} = (-\dot{q}_2 s_3 \;\; -\dot{q}_2 c_3 \;\; \dot{q}_3)^T \Rightarrow \frac{1}{2}\omega^T I_3 \omega^T = \begin{cases} \dot{q}_2^2 s_3^2 I_{3x} + \\ \dot{q}_2^2 c_3^2 I_{3y} + \\ \dot{q}_3^2 I_{3z} \end{cases}$$

$$^0T_2 = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & -1 & 0 & q_1 \\ 0 & 0 & 0 & 1 \end{pmatrix}\begin{pmatrix} c_2 & 0 & -s_2 & 0 \\ s_2 & 0 & c_2 & 0 \\ 0 & -1 & 0 & l_2 \\ 0 & 0 & 0 & 1 \end{pmatrix} = \begin{pmatrix} c_2 & 0 & -s_2 & 0 \\ 0 & -1 & 0 & l_2 \\ -s_2 & 0 & -c_2 & q_1 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

$$p_{c3} = \begin{pmatrix} c_2 & 0 & -s_2 & 0 \\ 0 & -1 & 0 & l_2 \\ -s_2 & 0 & -c_2 & q_1 \\ 0 & 0 & 0 & 0 \end{pmatrix}\begin{pmatrix} d_{c3}c_3 \\ d_{c3}s_3 \\ 0 \\ 1 \end{pmatrix} = \begin{cases} d_{c3}c_2c_3 \\ l_2 - d_{c3}s_3 \\ q_1 - d_{c3}s_2c_3 \end{cases} \Rightarrow v_{c3} = \begin{cases} -d_{c3}s_2c_3\dot{q}_2 - d_{c3}c_2s_3\dot{q}_3 \\ -d_{c3}c_3\dot{q}_3 \\ \dot{q}_1 - d_{c3}c_2c_3\dot{q}_2 + d_{c3}s_2s_3\dot{q}_3 \end{cases}$$

$$T_3 = \frac{1}{2} m_3 \left(d_{c3}^2 c_3^2 \dot{q}_2^2 + d_{c3}^2 \dot{q}_3^2 + \dot{q}_1^2 - 2d_{c3}c_2c_3\dot{q}_1\dot{q}_2 + 2d_{c3}s_2s_3\dot{q}_1\dot{q}_3\right)$$
$$\frac{1}{2}\left(\dot{q}_2^2 s_3^2 I_{3x} + \dot{q}_2^2 c_3^2 I_{3y} + \dot{q}_3^2 I_{3z}\right)$$

$e_1 = m_1 + m_2 + m_3 \quad e_2 = d_{c3} m_3 \quad e_3 = I_{2y} \quad e_4 = m_3 d_{c3}^2 + I_{3y}$

$e_5 = I_{3x} \quad e_6 = m_3 d_{c3}^2 + I_{3z}$

$$M_{11} = m_1 + m_2 + m_3 = e_1 \qquad M_{12} = -m_3 d_{c3} c_2 c_3 = -e_2 c_2 c_3 \qquad M_{13} = e_2 s_2 s_3$$

$$M_{22} = I_{2y} + m_3 d_{c3}^2 c_3^2 + I_{3x} s_3^2 + I_{3y} c_3^2 = I_{2y} + (m_3 d_{c3}^2 + I_{3y}) c_3^2 + I_{3x} s_3^2 \Rightarrow$$

$$M_{22} = e_3 + e_4 c_3^2 + e_5 s_3^2 \qquad M_{33} = m_3 d_{c3}^2 + I_{3z} = e_6$$

In M_{22} i used $S_3^2 = 1 - C_3^2$ and reduced the number of coefficients.

$$M(q) = \begin{bmatrix} e_1 & -e_2 c_2 c_3 & e_2 s_2 s_3 \\ -e_2 c_2 c_3 & e_3 + e_4 c_3^2 & 0 \\ e_2 s_2 s_3 & 0 & e_5 \end{bmatrix} =$$

$$p_{c_1 z} = q_1 \Rightarrow U_1 = m_1 g_0 q_1$$
$$p_{c_2 z} = q_1 \Rightarrow U_2 = m_2 g_0 q_1$$
$$\Rightarrow U = g_0 \left[m_3 (q_1 - d_{c3} s_2 c_3) + q_1 (m_1 + m_2) \right]$$

$$g_1 = g_0 e_1 \qquad g_2 = -g_0 m_3 d_{c3} c_2 c_3 = -g_0 e_2 c_2 c_3$$
$$g_3 = g_0 m_3 d_{c3} s_2 s_3 = g_0 e_2 s_2 s_3$$

$$\frac{\partial M_1}{\partial q} = \overset{\tilde{C}_1 =}{\begin{pmatrix} 0 & 0 & 0 \\ 0 & e_2 s_2 c_3 & e_2 c_2 s_3 \\ 0 & e_2 c_2 s_3 & e_2 s_2 c_3 \end{pmatrix}} \Rightarrow c_1 = \dot{q}^T \tilde{C}_1 \dot{q} = e_2 s_{23} (\dot{q}_2 + \dot{q}_3)$$

$$\frac{\partial M_2}{\partial q} = \begin{bmatrix} 0 & e_2 s_2 c_3 & e_2 c_2 s_3 \\ 0 & 0 & -2 e_4 s_3 c_3 \\ 0 & 0 & 0 \end{bmatrix} \qquad \frac{\partial M}{\partial q_2} = \begin{bmatrix} 0 & e_2 s_2 c_3 & e_2 c_2 s_3 \\ e_2 s_2 c_3 & -2 e_4 s_3 c_3 & 0 \\ e_2 c_2 s_3 & 0 & 0 \end{bmatrix}$$

$$\tilde{C}_2 = e_4 s_3 c_3 \begin{pmatrix} 0 & 0 & 0 \\ 0 & 1 & -1 \\ 0 & -1 & 0 \end{pmatrix} \Rightarrow c_2 = e_4 s_3 c_3 (\dot{q}_2 - 2\dot{q}_2 \dot{q}_3)$$

$$\frac{\partial M_3}{\partial q} + \frac{\partial M_3}{\partial q}^T = \begin{pmatrix} 0 & e_2 c_2 s_3 & e_2 s_2 c_3 \\ e_2 c_2 s_3 & 0 & 0 \\ e_2 s_2 c_3 & 0 & 0 \end{pmatrix} \qquad \tilde{C}_3 = e_4 s_3 c_3 \begin{pmatrix} 0 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 0 \end{pmatrix} \Rightarrow c_3 = -e_4 s_3 c_3 \dot{q}_2^2$$

$$\begin{cases} e_1 \ddot{q}_1 - e_2 c_2 c_3 \ddot{q}_2 + e_2 s_2 s_3 \ddot{q}_3 + e_2 s_{23} (\dot{q}_2 + \dot{q}_3) + g_0 e_1 = \tau_1 \\ -e_2 c_2 c_3 \ddot{q}_1 + (e_3 + e_4 c_3^2) \ddot{q}_2 + e_4 s_3 c_3 (\dot{q}_2 - 2\dot{q}_2 \dot{q}_3) - g_0 e_2 c_2 c_3 = \tau_2 \\ e_2 s_2 s_3 \ddot{q}_1 + e_5 \ddot{q}_3 - e_4 s_3 c_3 \dot{q}_2^2 + g_0 e_2 s_2 s_3 = \tau_3 \end{cases}$$

$$Y = \begin{bmatrix} \ddot{q}_1 + g_0 & s_{23}(\dot{q}_2 + \dot{q}_3) + s_2 s_3 \ddot{q}_3 - c_2 c_3 \ddot{q}_2 & 0 & 0 & 0 \\ 0 & -c_2 c_3 \ddot{q}_1 - g_0 c_2 c_3 & \ddot{q}_2 & \ddot{q}_2 c_3^2 + s_3 c_3 (\dot{q}_2 - 2\dot{q}_2 \dot{q}_3) & 0 \\ 0 & s_2 s_3 \ddot{q}_1 + g_0 s_2 s_3 & 0 & -s_3 c_3 \dot{q}_2 & \ddot{q}_3 \end{bmatrix}$$

the irrelevant standard parameters are 22, we show the relevant:

$m_1, m_2, m_3, d_{c3y} m_3, I_{2yy}, I_{3xx}, I_{3yy}, I_{3zz}$

Exercise 2

The 2R planar robot in Fig. 2 has equal links of length l and is commanded by the joint acceleration $\ddot{\boldsymbol{q}} \in \mathbb{R}^2$. The robot end effector has to perform a one-dimensional trajectory task $r_d(t) \in \mathbb{R}$ specified *only* along the x-direction. In a given robot state $(\boldsymbol{q}, \dot{\boldsymbol{q}})$, a desired task acceleration $\ddot{r}_d$ is assigned.

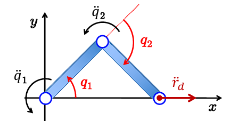


Figure 2: A 2R planar robot in a one-dimensional task.

Provide in symbolic form the command $\ddot{\boldsymbol{q}}$ that executes the task while minimizing the cost

$$H = \frac{1}{2}\left(\ddot{\boldsymbol{q}} - \ddot{\boldsymbol{q}}_0\right)^T\left(\ddot{\boldsymbol{q}} - \ddot{\boldsymbol{q}}_0\right), \qquad \ddot{\boldsymbol{q}}_0 = -\boldsymbol{K}_v\,\dot{\boldsymbol{q}},$$

for a diagonal matrix $\boldsymbol{K}_v > 0$. Evaluate then numerically the solution when the robot is in the configuration $\boldsymbol{q} = (\pi/4, -\pi/2)$ [rad] with joint velocity $\dot{\boldsymbol{q}} = (1, -1)$ [rad/s], the link length is $l = 1$ m, the task acceleration is $\ddot{r}_d = 1$ m/s^2, and $\boldsymbol{K}_v = \text{diag}\{2, 2\}$ [s^{-1}]. How would you modify the acceleration command if there was a task error in position and/or velocity?

The task is $r = l(c_1 + c_{12})$ hence the Jacobian is

$J = (-ls_1 - ls_{12} \quad -ls_{12})$. We want to solve the following opt. problem:

$$\begin{cases} \min\limits_{\ddot q} \frac{1}{2}(\ddot q - \ddot q_0)^T I (\ddot q - \ddot q_0) \\ \text{s.t. } J\ddot q = \ddot r_d - \dot J \dot q \end{cases}$$

this is a simple quadratic problem and have an analytical solution.

$$\ddot q = -K_v \dot q + J^T (JJ^T)^{-1} (\ddot r_d - \dot J \dot q - J(-K_v \dot q))$$

Now i substitute the numerical values: $c_1 = s_1 = \sqrt{2}/2$ $\quad c_{12} = \sqrt{2}/2$ $\quad s_{12} = -\sqrt{2}/2$

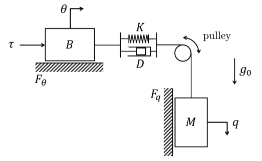
$$\dot J = \left(-lc_1\dot q_1 - lc_{12}(\dot q_1 + \dot q_2) \quad -lc_{12}(\dot q_1 + \dot q_2)\right) = \left(-\sqrt{2}/2 \quad 0\right)$$

$$J = (0 \quad \sqrt{2}/2) \qquad J^\# = (0 \quad \sqrt{2})^T \qquad -K_v \dot q = (-2 \quad 2)^T$$

$$\ddot q = \begin{pmatrix} -2 \\ 2 \end{pmatrix} + \begin{pmatrix} 0 \\ \sqrt{2} \end{pmatrix}\left(1 + \frac{\sqrt{2}}{2} - \sqrt{2}\right) = \begin{cases} -2 \\ 1 + \sqrt{2} \end{cases}$$

Exercise 3

Figure 3 shows a mechanical system made of two masses B and M connected by a pulley and a damped elastic spring, viscous friction on the motion of the individual masses, an input force τ acting on the first mass, and gravity acting on the second mass only. The zero of the two position variables θ and q is associated to an undeformed spring. The spring has stiffness $K > 0$ and its elastic potential energy is quadratic in the deformation $q - \theta$.



a) Derive the dynamic model of this system, including all non-conservative terms due to viscous friction (with coefficients $F_\theta > 0$ and $F_q > 0$, respectively) affecting the motion of the two masses and damping (with coefficient $D \geq 0$) on the time-varying deformation of the spring.

b) Provide the simplest feedback law that is able to asymptotically stabilize the position of the mass M to a constant desired height q_d. Prove the result using any preferred analysis method (linearization by Taylor expansion, Lyapunov/LaSalle, etc.).

c) Set now $D = 0$. Solve the inverse dynamics problem for a desired, sufficiently smooth trajectory $q_d(t)$. Provide the explicit expression of $\tau_d(t)$ as a function of $q_d(t)$ and its (higher order) time derivatives only.

The kinetic energy is: $\frac{1}{2}B\dot{\theta}^2 + \frac{1}{2}M\dot{q}^2$. The potential is

$$U = Mg_0 q + \frac{1}{2}K(\theta - q)^2$$

in. mat: $\begin{pmatrix} B & 0 \\ 0 & M \end{pmatrix}$ $\quad g = \begin{cases} K(\theta - q) \\ -K(\theta - q) - Mg_0 \end{cases} \Rightarrow \begin{cases} B\ddot{\theta} + K(\theta - q) = -D(\dot{\theta} - \dot{q}) - F_\theta \dot{\theta} + \tau \\ M\ddot{q} - K(\theta - q) - Mg_0 = D(\dot{\theta} - \dot{q}) - F_q \dot{q} \end{cases}$

We have to move θ in order to set $q = q_d$ while compensating gravity.

$$K(\theta - q) + Mg_0 = 0 \Rightarrow \theta = q - \frac{M}{K}g_0$$ so we should regulate θ to $\overbrace{q_d - \frac{M}{K}g_0}^{\theta_d}$

$\Rightarrow \tau = K_p\left(q_d - \frac{M}{K}g_0 - \theta\right)$ I want to show that $\underset{x_e}{(\theta_d, q_d, 0\ 0)}$ is an equilibrium state.

const

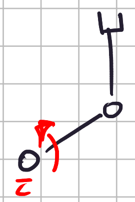
$$V = \frac{1}{2}B\dot{\theta}^2 + \frac{1}{2}M\dot{q}^2 + Mg_0 q + \frac{1}{2}K(\theta - q)^2 + K_p(\theta_d - \theta) + U_0$$

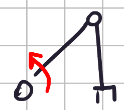
$$\dot{V} = \dot{\theta}B\ddot{\theta} + \dot{q}M\ddot{q} + Mg_0\dot{q} + K(\theta - q)(\theta - q) - K_p\dot{\theta}$$

Exercise 4

Suppose that a 2R planar robot that is moving in a vertical plane has only one actuator at the first joint providing a torque τ. Find the expression of all forced equilibria $(\bar{\boldsymbol{q}}, \mathbf{0})$ associated to a generic constant input torque $\bar{\tau}$. Are you able to find a state feedback control law $\tau = f(\bar{\boldsymbol{q}}, \boldsymbol{q}, \dot{\boldsymbol{q}})$ that asymptotically stabilizes one of these equilibria, at least locally?

$$\begin{cases} (e_1+2e_2c_2)\ddot{q}_1 + (e_3+e_2c_2)\ddot{q}_2 - e_2s_2(\dot{q}_2^2+\dot{q}_1\dot{q}_2) + e_4c_1 + e_5c_{12} = \tau \\ (e_3+e_2c_2)\ddot{q}_1 + e_3\ddot{q}_2 + e_2s_2q_2\dot{q}_1 + e_5c_{12} = 0 \end{cases}$$





For the forced equilibria we set $\dot{q} = \ddot{q} = 0$

$$\begin{cases} e_4c_1 + e_5c_{12} = \bar{\tau} \\ e_5c_{12} = 0 \end{cases} \Rightarrow q_1+q_2 = \pm\frac{\pi}{2} \Rightarrow e_4c_1 = \bar{\tau} \Rightarrow c_1 = \frac{\bar{\tau}}{e_4} \Rightarrow q_1 = \arccos\left\{\frac{\bar{\tau}}{e_4}\right\}$$